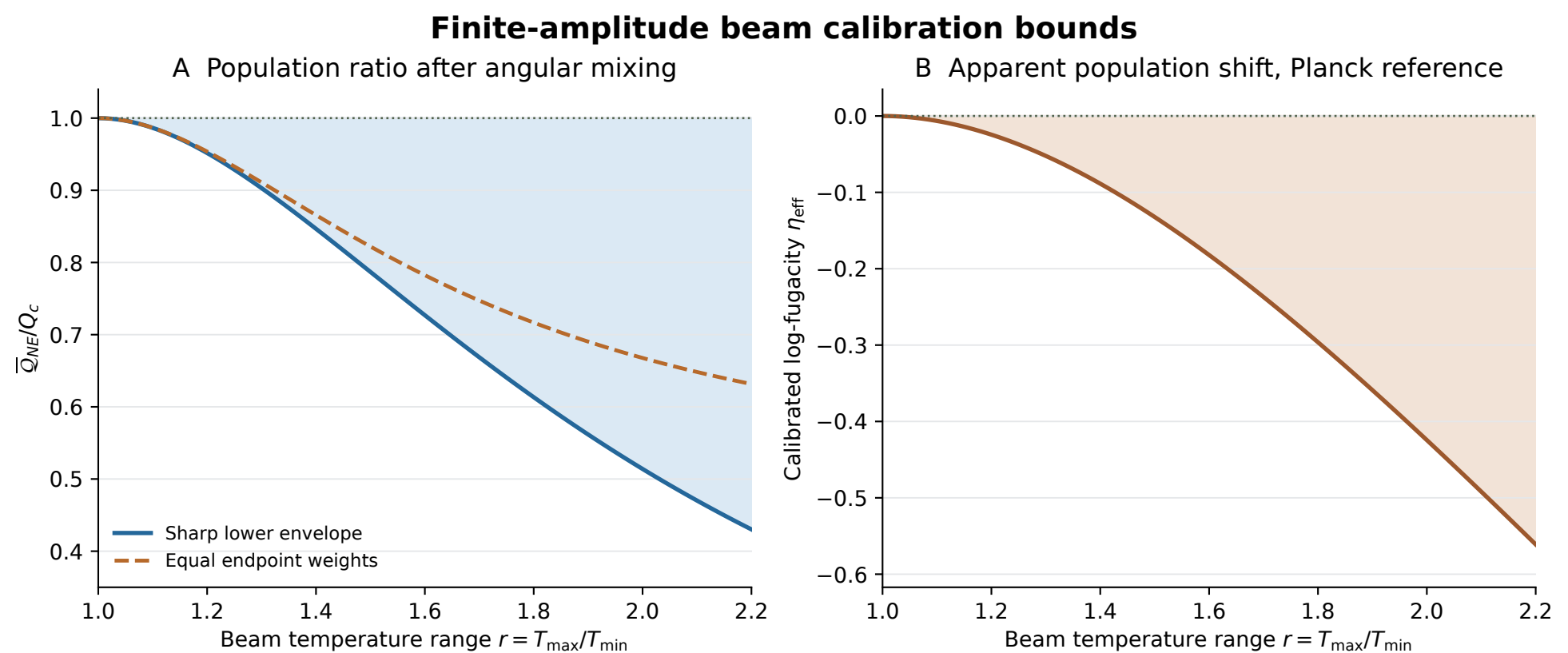
Finite-amplitude beam calibration bounds
A Population ratio after angular mixing
B Apparent population shift, Planck reference
$\overline{Q}_{NE}/Q_c$
Calibrated log-fugacity $\eta_{\mathrm{eff}}$
Sharp lower envelope
Equal endpoint weights
Beam temperature range $r = T_{\max}/T_{\min}$
Beam temperature range $r = T_{\max}/T_{\min}$
Positive, energy-independent beam; all underlying rays have η = 0. Calibrated η is a two-moment representative, not a true chemical distortion.